# Offensive Security Engineering

| Level | Product Output | Decision Making / Supervision | Communication / Writing | Collaboration | Adversary Emulation and Tradecraft | Application and Cloud Security | Research and Tooling |
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| 1 | Applies specified techniques to well-scoped systems in order to achieve previously defined objectives. | Normally receives general instruction on work and new assignments. | Reports progress on a regular basis. Documents technical issues clearly and accurately. Actively seeks assistance as needed. | Works with peers to make progress on well-scoped tasks as part of a larger engagement. | Applies established techniques and tools to execute a defined adversary scenario.<br><br>Collects evidence and explains findings regarding controls and detections. | Applies well-established application and cloud security fundamentals when reviewing clearly scoped systems and configurations.<br><br>Identifies security weaknesses and their implications in terms of specific threats. | Reads security research to learn from others.<br><br>Attempts to replicate vulnerabilities and attacks. |
| 2 | Chooses techniques to apply to well-scoped systems in order to achieve previously defined objectives. | Normally receives little instruction on day-to-day work, general instructions on new assignments. | Effectively provides critical feedback to peers. Writes clear, reproducible findings. | Works independently alongside peers to accomplish well-scoped tasks as part of a larger engagement. | Selects relevant TTPs to emulate and explains what the results demonstrate in the context of specific threat actors or risks. | Reasons about how a system's behavior, identity, permissions, and configuration enforce intended security boundaries.<br><br>Traces weaknesses to underlying design and implementation decisions. Develops remediation recommendations that take into account intended functionality and requirements. | Establishes how published research applies to Teleport's systems.<br><br>Adapts published techniques to our needs.<br><br>Writes scripts and internal tools to conduct attacks and facilitate testing. |
| 3 | Chooses scenarios and techniques that emulate relevant threats in order to achieve defined security objectives in multiple related systems. | Work is done independently and is reviewed at critical points. | Clearly communicates project scope, limitations, threat models. | Collaborates on a short feedback loop for detection engineering strategies through automated and manual processes. | Translates existing relevant threats and defined security objectives into exercise plans with specific hypotheses about prevention and detection.<br><br>Selects scenarios and decides the evidence needed to establish the effectiveness of controls. | Evaluates security controls across multiple related systems and connected infrastructure. Identifies weaknesses resulting from incorrect assumptions about controls.<br><br>Chains multiple identified weaknesses to demonstrate their combined security impact. | Develops tools for security tasks relevant to Teleport and makes them useable by other members of the team. |
| 4 | Determines exercise goals and targets based on current threats and new risks to the organization.<br><br>Connects findings to tangible hardening recommendations with specific suggested remediations. | Work is reviewed upon completion and is consistent with departmental objectives. | Communicates impact of work through well structured, easily understood reporting to relevant parties. | Provides strong technical recommendations for hardening systems backed by clearly documented attack simulation results. | Identifies and discovers relevant threats and defines security objectives based on system architecture and gaps in understanding of detective controls.<br><br>Prioritizes exercises to resolve uncertainty about critical defensive controls and detection strategies. | Develops well-researched architectural recommendations that address security weaknesses across multiple systems working together.<br><br>Develops alternative designs and explains their security and engineering tradeoffs. | Investigates security problems without an established answer.<br><br>Publishes blog posts externally regarding their research work relevant to the organization.<br><br>Makes tooling useful to external security researchers. |
| 5 | Builds and maintains the process for selection, development, execution of advanced simulation exercises.<br><br>Recognized as an expert in offensive security internally and externally. | Focuses on providing thought leadership and works on broader organization projects, which requires understanding of wider business. Recognized internally as a subject matter expert. May direct the activities of others. | Educates colleagues on threats, attack scenarios, and defensive controls.<br><br>Writes both internal and external facing technical and strategic content.<br><br>Develops helpful frameworks for communicating complex concepts. | Influences the engineering roadmap through clear demonstration of risk accompanied by technically sound proposals for remediation and hardening.<br><br>Influences detection engineering roadmap through the establishment of collaborative processes. | Validates operational effectiveness, identifies gaps, and develops practices that improve the quality of the team's work over time.<br><br>Uses threat intelligence and published research to identify important gaps in the organization's understanding of adversaries. | Continuously develops expertise in new platforms and technologies relevant to the organization. | Presents their work at conferences relevant to the organization. |
| 6 | Produces novel operational capabilities and research. | Wide latitude in determining objectives and approaches to critical assignments. | Documents complex novel techniques and operational capabilities to both internal and external parties. | Collaborates with industry peers to advance the capabilities of the broader security community. | | | Develops novel techniques against hardened targets that advance the state of the industry. |